

# Readable, Faithful, Used: Three Dissociable Properties of Demographic Identity in a Language Model

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## Abstract

Large language models are widely used to simulate survey respondents, yet their outputs are homogeneous and unfaithful to real inter-group differences, and whether this reflects what a model knows or uses has remained untested. Using representational similarity analysis against Pew American Trends Panel ground truth, we score demographic read-out locations in Mistral-7B and intervene causally across six attribute types. The internal geometry is faithful: attention-head read-outs dominate the standard residual read-out, reaching selection-corrected  $\rho$  up to 0.63—about 70% of the measurement-reliability ceiling—and one head, L11 H16, is significantly faithful across all six types, though race-based types stay weak and prompt-fragile, replicating in a second model family. Yet causal use does not track fidelity: the clearest causal pathway ( $p = 0.002$ ) sits in one of the least faithful types, the most faithful type shows no correction-surviving effect, and full identity swaps in the prompt move predictions by under 2% of their error. A 128-dimensional probe on that head lands 21–31% closer to survey truth than the model’s answers, yet recovers almost none of the per-question group ordering. Readable, faithfully arranged, and causally used are three dissociable properties of the same model; treating them as one claim is what keeps the “can LLMs simulate populations” debate unresolved.

## 1 Introduction

LLMs are increasingly deployed as simulated survey respondents and human simulators [29, 37]. Yet behavioral evidence shows their outputs are excessively uniform: simulated users are “overly agreeable” and “stylistically uniform” [38], opinion predictions barely differentiate demographic groups, and general model capability does not predict simulation fidelity [38]. This evidence is *behavioral*—measured from the outside. Whether the failure lives in what the model *knows* or in what it *uses* has remained open, because the two research communities each hold only half of the required instrumentation.

Interpretability work localizes demographic and persona information inside models—down to individual attention heads [15, 17]—but validates against the model’s own behavior (probe accuracy, steering success), never against external population ground truth. LLM-survey-simulation work has exactly that ground truth (real per-group response distributions) but treats the model as a black box. The experiment that requires both instruments at once—whether the internal arrangement of groups matches how those groups actually answer—sits in the gap between them and had not been run.

We hold both instruments at once. For 169 intersectional demographic cells (e.g., *Asian*  $\times$  *Hindu*, *age 18–29*  $\times$  *Democrat*) we compute two representational dissimilarity matrices—one from

real Pew ATP response distributions (Wasserstein distances between groups’ answers), one from the model’s internal representations at a given location—and correlate them (RSA; 18). Applied at  $1,089 \text{ locations} \times 4 \text{ prompt operationalizations}$ , this yields a *fidelity map*: where inside the model does the arrangement of demographic groups mirror how the groups actually differ in opinion?

Our results split the question “can LLMs simulate populations?” into three separable properties (we reserve “layer” for transformer layers):

1. **Readable** (established by prior work): per-group opinion distributions can be decoded from internal activations [15].
2. **Faithfully arranged** (this work): at specific locations, including a single attention head, L11 H16, the *inter-group geometry* mirrors real opinion structure (selection-corrected  $\rho$  up to 0.63), which per-group readability neither implies nor requires.
3. **Actually used** (this work): the faithfully arranged information barely flows into the model’s answers—full identity swaps in the prompt move predictions by  $<2\%$  of their error, and not directionally toward the target group’s truth. A strong-instrument intervention does find a real causal pathway, but where it is clearest is *not* where the map is most faithful (Sec. 5.3).

## Contributions.

- A *fidelity map* of demographic identity in an LLM—to our knowledge the first to score internal inter-group geometry against real survey ground truth: RDM-vs-RDM comparison across layers, attention heads, FFN outputs, and prompt variants, with winner’s-curse-corrected statistics (max-statistic permutation, held-out and split-half validation)—and identifies L11 H16 as a general demographic-geometry head, significant across all six attribute types as a *fixed* location, with fidelity heterogeneous by type: political/socio-economic structure is encoded at  $\rho$  up to 0.63, while racial $\times$ religious structure is weak and prompt-fragile everywhere.
- A dissociation of causal use from representational fidelity. Replacing the entire prompt identity moves the model’s predicted distribution by under 2% of its prediction error in aggregate (Sec. 5), and a causal sweep over all 32 layers in all six types—with cluster-robust, selection-corrected inference—shows that the clearest causal pathway sits in a low-fidelity type while the most faithful type shows no correction-surviving single-layer pathway (Sec. 5.3).
- A direct measurement of what a faithful read-out buys, with a practical reframing. A probe on L11 H16 alone (128 dimensions) is 21–31% closer to survey truth than the model’s own answers, but recovers almost no per-question group ordering: the two measure different resolutions, faithful *on average* but not sharp enough for any single question (Sec. 5.4). If faithful structure already exists internally, surface fine-tuning [29] may re-teach what the model already encodes—a candidate mechanism for its known transfer failure—and reading the location directly is the cheaper alternative, though our probe results show it is still not a substitute for survey data.

## 2 Related Work

**LLMs as survey simulators (the ground-truth side).** “Silicon sampling” was introduced by Argyle et al. [1] and has since been contested: response instability and survey-format sensitivity [8, 24], systematic divergence from human samples [3], and bounded persona effects [14]. SubPOP

[29] fine-tunes LLMs to predict per-group response distributions with a KL objective, with strong in-distribution but degraded out-of-survey accuracy; rectification methods [19] combine limited human data with synthetic responses. Cultural-alignment work compares model *outputs* to cross-national survey structure—GlobalOpinionQA against World Values Survey and Pew Global Attitudes items [9], and ChatGPT responses against Hofstede’s dimensions [6]. All treat the model as a black box.

**Reading opinions from inside (the closest neighbor).** Jahanparast et al. [15] show linear probes on internal activations recover per-group opinion distributions *better than the model’s own generated outputs*, extract activations down to attention heads, and steer outputs via SAE patching. All their measurement is *first-order*: each group’s read-out is scored against that group’s own distribution. Whether the *inter-group geometry* at any location mirrors real inter-group opinion structure—the property generalization depends on—is not among their measurements.<sup>1</sup> Our work is the second-order complement: probe accuracy is a phone book, accurate per entry; we ask whether the entries are *arranged as a map*. Two works come closest: Bhattacharyya et al. [2] correlate persona-conditioned final-layer separation with human subgroup disagreement, and Williams et al. [36] test whether aligned models reproduce human item–item correlation beyond marginals—both compare behavioral or last-layer structure, not internal geometry against a human inter-group distance matrix. Layer-wise RSA of LLM geometry against human similarity structure exists for *perceptual* domains, peaking mid-stack and fading with depth [28]—a parallel to our demographic-identity finding.

**Localizing social attributes (the interpretability side).** Kim et al. [17] localize political ideology to attention heads and validate against legislator vote-based scores (DW-NOMINATE)—our closest methodological relative (one attribute, elite text); a follow-up locates a single partisan axis mid-stack and steers along it [30]. Cintas et al. [7], Bouchaud and Ramaciotti [4], Poonia and Jain [23] localize persona and socio-demographic representations—via linear probing across layers and activation patching—without survey ground truth. Per-layer RSA applies to LLMs in other content domains [27]. “Diversity collapse” [32] documents social-identity representations becoming less separable with depth—consistent with our anisotropy findings—but not fidelity to real opinion structure. Construct-validity critiques of persona prompting [31] motivate our multi-cue design. Gilg et al. [12] find preference machinery largely shared across personas—probes and steering vectors transfer between opposed personas—converging with our dissociation between what a location encodes and what the model answers with.

**The behavioral Sim2Real gap (the motivation side).** The Sap-group program documents from outside that LLM simulators are homogeneous and overly cooperative: 451-human comparisons show the best of 31 simulators reaches only USI 76.0 vs. a 92.7 human baseline [38], and OdysSim [37] attributes the gap to helpfulness-driven post-training and rebuilds a behavioral foundation model to close it—noting that better *average* human simulation does not imply faithful *subpopulation* simulation. Persona effects on moral stances are strongest for political ideology and personality [20]. Foundational annotation work [10, 26] established that human judgments are distributions conditioned on who answers—our evaluation’s premise. None of this line opens the model.

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<sup>1</sup>Their four reported methods—linear probing, SAE feature analysis, SAE-based steering, and fine-tuning—are all first-order, per-group comparisons (ICLR 2026 camera-ready, OpenReview [kHVzEjThKE](#); repository [schang-lab/llm-opinions](#)); no inter-group RSA or distance-matrix analysis is described.

### 3 Method

#### 3.1 Survey-side ground truth

We use OpinionQA-style per-group response distributions derived from Pew American Trends Panel waves [25, 29] (15 waves; inventory and preprocessing in Appendix A.1). A *group* is an intersectional cell: a pair of demographic attribute values (e.g., *race=Asian*  $\times$  *religion=Hindu*), yielding 169 cells across six attribute types (AGE $\times$ POLPARTY, EDUCATION $\times$ INCOME, RACE $\times$ RELIG, RACE $\times$ POLPARTY, RACE $\times$ POLIDEOLOGY, RELIG $\times$ POLPARTY). These six pairs are a purposive, not exhaustive, sample of the  $\binom{12}{2} = 66$  possible attribute pairs in this corpus, chosen to instantiate an intersectional dimension prior work flags as an open question and to span distinct social/political axes (full justification in Appendix A.2). By number of cells clearing the  $n \geq 30$  threshold, our six rank 7th to 44th among all 66 pairs, so selection was not biased toward the best-populated ones (Appendix A.2). The pairs were fixed, and the underlying cell data built, before any fidelity or causal result was computed on them. For groups  $i, j$  the ground-truth distance is the mean Wasserstein distance between their answer distributions over shared questions (up to 300 sampled per pair):

$$D_{ij}^{\text{real}} = \frac{1}{|Q_{ij}|} \sum_{q \in Q_{ij}} \text{WD}(p_i^{(q)}, p_j^{(q)}), \quad (1)$$

where  $p_i^{(q)}$  is group  $i$ ’s real response distribution on question  $q$ ’s ordinal scale.  $D^{\text{real}}$  is the survey-side RDM. Appendix B.7 reports sensitivity of the headline results to two construction choices: normalising WD by each question’s scale range, and restricting to questions answered by every cell of a type.

#### 3.2 Model-side representations and prompts

For each group we build identity prompts and extract, at the final token, (i) the residual stream after every layer (33 locations), (ii) each attention head’s output before the out-projection ( $32 \times 32 = 1,024$  locations; the same read point used by 15), and (iii) each FFN block output (32 locations), from Mistral-7B [16]—our primary model throughout. The same read-out protocol is reapplied unchanged to a second model family in Sec. 4.6 (Qwen3-8B-Base and two OdysSim checkpoints, whose 36-layer stack yields  $37 + 1,152 + 36 = 1,225$  locations by the same construction) and to Mistral-7B-Instruct-v0.2 for the identity-swap ceiling in Sec. 5.1. The causal experiments of Sec. 5 use activation patching [21, 33], following Heimersheim and Nanda [13] and mindful of documented reliability limits [5]. The model-side RDM at location  $\ell$  uses cosine distances between group representations.

A single prompt template is a single operationalization of “group identity” [31]. We therefore use four templates per group (third-person declarative, first-person, structured profile, and QA format ending in **Answer:**—the read-out position validated by 15) and additionally a template-averaged representation (TMEAN); the four are reproduced verbatim in Appendix A.3. Template disagreement is reported as a fragility diagnostic.

#### 3.3 Fidelity: second-order comparison

Fidelity at location  $\ell$  for attribute type  $t$  is the Spearman correlation between the upper triangles of the two RDMs restricted to type- $t$  cells:

$$\rho_{\ell,t} = \text{Spearman}\left(\{D_{ij}^{\text{real}}\}_{i < j \in t}, \{D_{ij}^{\ell}\}_{i < j \in t}\right). \quad (2)$$

We report within-type fidelity throughout: pooled correlations mix between-type scale differences and can both mask weak types and manufacture spurious gains (Sec. 4.1).

### 3.4 Causal interventions and probes

Beyond representational geometry, we test whether outputs causally depend on it, with four instruments (Appendix C.1; Appendix D.1). An identity-swap ceiling measures how much the predicted opinion distribution moves when the entire demographic identity is replaced, bounding how identity-responsive the output is at all. Activation patching at the identity token replaces a donor group’s activations there, propagating the edit forward; our primary intervention jointly patches all 32 heads of layer 11. A per-layer sweep repeats this 32-head patch at every layer for all six types; significance comes from a max-statistic sign-flip permutation test over layers, computed at the pair level because items cluster into donor–recipient pairs. Finally, a linear probe fit on L11 H16’s 128 activations (leave-one-cell-out) asks how much of a group’s real response distribution is linearly recoverable from that location alone.

### 3.5 Statistical protection against selection

Any “best location” claim over 1,089 candidates inflates under selection (winner’s curse). We therefore report three corrections: (a) **max-statistic permutation tests** [22]—the null distribution of the *maximum*  $\rho$  over all locations under cell-label permutation; (b) **held-out-template selection**—locations chosen on three templates, scored on the fourth; (c) **split-half selection**—locations chosen on half the cells, scored on the other half (200 resamples). Headline numbers are held-out values, not selected maxima.

Two further disciplines apply to the causal experiments (Sec. 5). Patching items are *clustered*: each type contributes 12 donor–recipient pairs  $\times$  20 questions, and items within a pair share the persona and its idiosyncrasies, so all causal  $p$ -values are computed at the *pair* level (sign-flip permutation over 12 clusters, enumerated exactly; Appendix C.2), with item-level statistics shown only as descriptive detail. And because the survey-side ruler could itself be confounded by lexical similarity of the identity words, Sec. 4 reports fidelity after partialling out a lexical-baseline RDM (Appendix B.5).

## 4 Results I: Where Demographic Geometry Is Faithful

### 4.1 The standard residual read-out understates the model

The read-out used implicitly across the LLM-survey literature—last-token residual stream, cosine distance, one prompt template—paints a bleak picture, but a misleading one. Pooled fidelity is  $\rho \approx 0.17$  (peak at layer 17), but this pools unequal types: AGE $\times$ POLPARTY and EDUCATION $\times$ INCOME reach  $\rho \approx 0.50$  while RACE $\times$ RELIG is  $\rho = 0.07$  (n.s.), and cross-type pairs correlate *negatively*, so distances are only meaningful within a type. Group embeddings are severely anisotropic ( $\|\bar{x}\|/\|x\| = 0.95$ ): mean-centering doubles the pooled correlation ( $0.185 \rightarrow 0.301$ ) yet *decreases* within-type correlation in all three types measured—a between-type scale artifact we do not adopt anywhere in this paper (Appendices B.1 and B.1). Top-5 neighbor precision against ground truth is  $\sim 2\times$  chance in all six types but only 0.35–0.48 absolute, and applications that borrow statistical strength from neighbors inherit this noise: a consistency-regularization pilot built on this read-out failed accordingly (Appendix E).

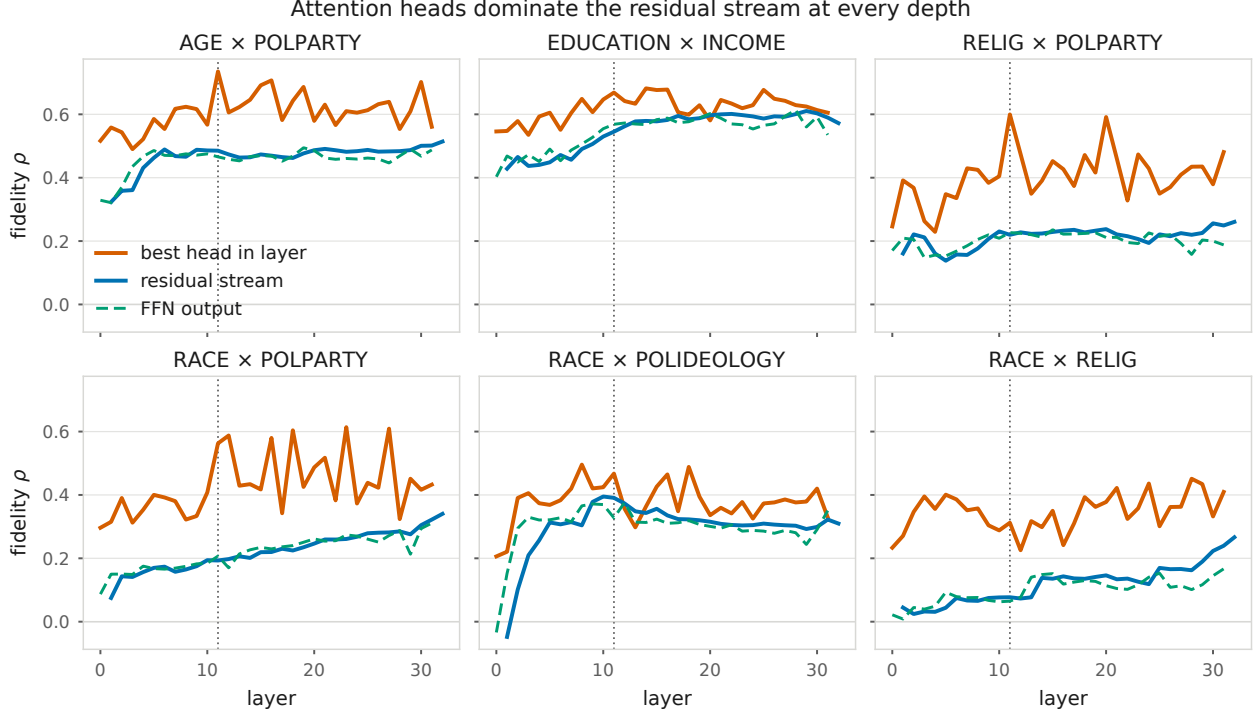


Figure 1: Fidelity by depth and component (multi-cue TMEAN read-out): orange = best of 32 heads per layer, blue = residual stream, dashed green = FFN output. Dotted line marks layer 11. Maxima here are selected/uncorrected; Table 1 gives held-out values.

Weakness of *one* read-out does not establish weakness of the *model*: the residual stream sums all attention heads and FFN blocks, so signal carried by a few components can be diluted by the rest. The next subsection reads every component separately.

## 4.2 Attention heads dominate the residual stream

The best attention-head read-out exceeds the best residual-stream read-out in every attribute type, with the largest gains where the standard read-out was weakest (RELIG×POLPARTY: 0.26 → 0.60; RACE×POLPARTY: 0.34 → 0.61; Table 1); under the symmetric correction of Sec. 4.3 this holds in five of six types, reversing in none. Multi-cue averaging alone already helps (RACE×RELIG residual: 0.07 → 0.27 TMEAN), so part of the apparent weakness was operationalization noise. Figure 1 shows this is not one lucky depth: the best head within a layer beats that layer’s residual stream at nearly every depth in all six types (residual wins at only 3 of 186 layer–type points), while the FFN output tracks the residual stream rather than the heads—consistent with dilution at the summation, not a failure to encode the signal.

Those per-layer curves take the *best* head at each depth and are selection-contaminated; averaging  $\rho$  over all 32 heads of a layer instead gives a selection-free view of which depths carry the structure (Table 10 in Appendix B.2). The first quarter of the stack is uniformly the weakest block, and the peak is not uniformly mid-stack: it emerges latest for RACE×RELIG, whose block means rise monotonically into the final quarter—the weakest type is thus also the slowest to build up.

Table 1: Fidelity ( $\rho$  vs. survey ground truth) by read-out. *Selected* = maximum over locations; *held-out* = selection-corrected range from two independent corrections (held-out template, Appendix B.3; split-half median), on the NaN-cleaned cell set.

| Type             | Residual best | Head best (selected) | Head (held-out) | Best loc. |
|------------------|---------------|----------------------|-----------------|-----------|
| AGE×POLPARTY     | +0.51         | +0.74                | +0.59–0.63      | L11 H19   |
| EDUCATION×INCOME | +0.61         | +0.68                | +0.58–0.63      | L14 H1    |
| RELIG×POLPARTY   | +0.26         | +0.60                | +0.50–0.52      | L11 H16   |
| RACE×POLPARTY    | +0.34         | +0.61                | +0.41–0.52      | L23 H13   |
| RACE×POLIDEOLOGY | +0.40         | +0.50                | +0.30–0.41      | L8 H21    |
| RACE×RELIG       | +0.27         | +0.45                | +0.21–0.32      | L28 H0    |

Table 2: Every control in Sec. 4, per attribute type.  $p_{\text{sel}}$ : selection-corrected max-statistic  $p$ . head/resid/proj: held-out fidelity of the three families (proj = random 128-d projections). % ceil: L11 H16 fidelity as a fraction of the attenuation ceiling. partial  $\rho$ : L11 H16 fidelity partialling out the lexical-baseline RDM.

| Type             | selection        | symmetric + geometry (held-out) |       |      | ceiling | lexical        |        |
|------------------|------------------|---------------------------------|-------|------|---------|----------------|--------|
|                  | $p_{\text{sel}}$ | head                            | resid | proj | % ceil  | partial $\rho$ | $p$    |
| AGE×POLPARTY     | <.0005           | 0.52                            | 0.39  | 0.42 | 74%     | 0.59           | <.0005 |
| EDUCATION×INCOME | <.0005           | 0.54                            | 0.51  | 0.49 | 70%     | 0.35           | <.0005 |
| RELIG×POLPARTY   | <.0005           | 0.44                            | 0.17  | 0.18 | 68%     | 0.59           | <.0005 |
| RACE×POLPARTY    | <.0005           | 0.47                            | 0.28  | 0.30 | 69%     | 0.58           | <.0005 |
| RACE×POLIDEOLOGY | .0025            | 0.38                            | 0.26  | 0.29 | 53%     | 0.37           | <.0005 |
| RACE×RELIG       | .0155            | 0.29                            | 0.16  | 0.17 | 39%     | 0.23           | .056   |

### 4.3 Controls: selection, geometry, and lexical confounds

Max-statistic permutation tests (Sec. 3.5) show pure noise picking its champion among 1,024 heads reaches only  $\rho \approx 0.34$ –0.46 (95th null percentile), exceeded by the observed maxima in all six types ( $p < 0.0005$  in four; 0.0025, 0.0155 in the rest). Held-out-template and split-half estimates agree (Table 1); head *choice* is stable under resampling for most types (L11 H16 re-selected in 87/200 splits for RELIG×POLPARTY) but a lottery for RACE×RELIG.

**Symmetric correction and a geometry control.** That comparison is also unequal in dimensionality: 128-dimensional heads might beat the anisotropic 4,096-dimensional stream for geometric, not informational, reasons. One pipeline with identical selection over three families—1,024 heads, 33 residual layers, and 1,024 random 128-dimensional projections of the residual stream (Appendix B.4)—shows heads winning in five of six types, with projections tracking the residual stream rather than the heads, so the advantage is not a dimensionality artifact; the exception, EDUCATION×INCOME, has a thin margin, where the standard read-out is already close to the best we find (Table 2).

**A ceiling for fidelity.** A correlation of 0.63 means something different under a ceiling of 0.70 than under 0.95. Survey-side split-half reliability is 0.985–0.994; model-side reliability across template halves at L11 H16 is 0.72–0.94, giving an attenuation ceiling  $\sqrt{r_{\text{survey}} r_{\text{model}}} = 0.84$ –0.96 per type (observed fidelity reaches 39–74% of it, Table 2): the unexplained residual is real signal the representation lacks, not an artifact of a noisy ruler.

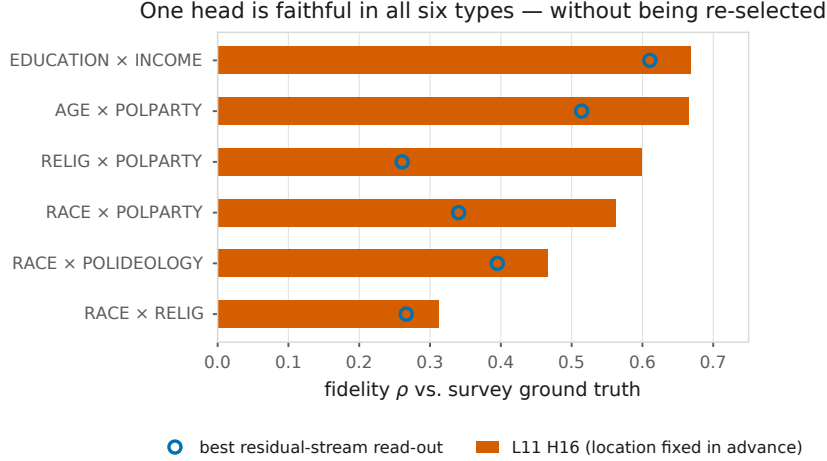


Figure 2: Fidelity of the fixed location L11 H16 (bars, never re-selected per type) against the best residual-stream read-out for that type (circles); margins  $+0.05$  to  $+0.34$ . Significance ( $p < 0.001$  each,  $\times 1024$  Bonferroni) is on the NaN-cleaned cell subset, where race-containing values shift slightly (RACE $\times$ RELIG  $0.31 \rightarrow 0.33$ ; RACE $\times$ POLPARTY  $0.56 \rightarrow 0.59$ ; exclusions in Appendix A.1).

**Not an artifact of the identity words themselves.** The prompts differ across cells only in the attribute-value words (“Democrat”, “Hindu”, “30–49”, ...), whose lexical similarity correlates with real opinion similarity, so the map could be a mere copy of word similarity. A baseline RDM built from a small sentence encoder [34] (value words alone, and the full identity sentence) is correlated with survey truth; we report each read-out’s *partial* Spearman correlation after controlling for it, by cell-label permutation.

The confound is real but does not explain the map: the baseline alone reaches  $\rho = 0.21$ – $0.64$  (largest for EDUCATION $\times$ INCOME, whose value words are ordered), yet L11 H16’s fidelity controlling for it remains  $\rho = 0.35$ – $0.59$  ( $p < 0.0005$ ) in five of six types, with the largest drop for EDUCATION $\times$ INCOME ( $0.67 \rightarrow 0.35$ ), consistent with its word-order structure. The exception is again RACE $\times$ RELIG: partial fidelity there is  $0.23$  ( $p = 0.056$ ; the type’s best head survives at  $0.41$ ,  $p = 0.0015$ )—the fourth control this type is weakest under (Table 2). For word-ordered types, though, some of the raw correlation is lexical.

#### 4.4 A general demographic-geometry head

Fixing L11 H16 in advance (no per-type re-selection) yields significant fidelity in *all six* types ( $\rho = 0.33$ – $0.67$ ;  $p < 0.001$  each, surviving a conservative  $\times 1024$  Bonferroni): a single head, one of 1,024, tracks real opinion structure across age, education, income, religion, party, ideology, and (weakly) race (Figure 2), and, being fixed before scoring, is structurally immune to the per-type winner’s curse of Sec. 3.5. One provenance caveat: L11 H16 was *noticed* for recurring in per-type top-10 lists, so this rules out per-type selection but not the initial noticing; independent confirmation comes from the causal test in Sec. 5 and the cross-model replication in Sec. 4.6, where the phenomenon recurs at a family-specific address (full head-by-head map in Appendix B.2).

Table 3: Cross-model replication of the fidelity map. *head/resid* = selected best head vs. best residual read-out.  $\rho_{\text{ho}}$  = held-out-template fidelity (max-statistic  $p$  in parentheses, 2,000 permutations). Mistral column: that model’s held-out head value, for reference.

| Type             | Mistral            | Qwen3-8B-Base |                            | OSim-8B-Mid |                            | OSim-8B (post-RL) |                            |
|------------------|--------------------|---------------|----------------------------|-------------|----------------------------|-------------------|----------------------------|
|                  | $\rho_{\text{ho}}$ | head/resid    | $\rho_{\text{ho}}$ ( $p$ ) | head/resid  | $\rho_{\text{ho}}$ ( $p$ ) | head/resid        | $\rho_{\text{ho}}$ ( $p$ ) |
| AGE×POLPARTY     | 0.52               | 0.78/0.50     | 0.49 (<.0005)              | 0.74/0.50   | 0.61 (<.0005)              | 0.73/0.51         | 0.57 (<.0005)              |
| EDUCATION×INCOME | 0.54               | 0.70/0.51     | 0.58 (<.0005)              | 0.68/0.50   | 0.54 (<.0005)              | 0.68/0.53         | 0.60 (<.0005)              |
| RELIG×POLPARTY   | 0.44               | 0.57/0.37     | 0.56 (<.0005)              | 0.58/0.39   | 0.56 (<.0005)              | 0.56/0.38         | 0.55 (<.0005)              |
| RACE×POLPARTY    | 0.47               | 0.60/0.37     | 0.37 (.001)                | 0.57/0.38   | 0.38 (.002)                | 0.63/0.37         | 0.46 (<.0005)              |
| RACE×POLIDEOLOGY | 0.38               | 0.59/0.35     | 0.38 (<.0005)              | 0.62/0.36   | 0.38 (<.0005)              | 0.60/0.35         | 0.40 (<.0005)              |
| RACE×RELIG       | 0.29               | 0.41/0.35     | 0.29 (.15)                 | 0.45/0.36   | 0.30 (.062)                | 0.43/0.33         | 0.22 (.096)                |

#### 4.5 Heterogeneity across attribute types

The type hierarchy is preserved at every location tested: the age- and education-based types are strongly and stably encoded (template std 0.02–0.05; RELIG×POLPARTY sits between, at 0.10), while every race-containing type is weaker *and* prompt-fragile (template std 0.12–0.13; e.g., RACE×RELIG at its best head:  $\rho = 0.16$  under one template, 0.45 under another). This parallels output-level findings that political ideology dominates persona effects [20] and annotator-attitude findings that ideological variables outpredict raw demographics [26]. Demographic attributes therefore cannot be treated as interchangeable: a persona pipeline validated on political or socio-economic attributes gives no warrant for race-based attributes, where this model’s internal structure is weaker and unstable under paraphrase.

#### 4.6 Cross-model replication

The sensitivity analyses above are internal checks; this subsection adds an external one: replicating the fidelity map on three checkpoints differing only in human-simulation training [37]—Qwen3-8B-Base, OSim-8B-Mid (midtrained on a 10B-token behavioral corpus), and OSim-8B (post-RL)—with the same prompts, cells, and pipeline, validated by reproducing the Mistral map from raw activations.

Four results: heads dominate the residual stream here too (best head beats best residual in all six types at all three checkpoints, tracking Mistral’s magnitudes, e.g. RELIG×POLPARTY 0.56 vs. 0.44); the type hierarchy replicates, race weakness included (five of six types pass max-statistic correction at every checkpoint,  $p \leq 0.0025$ ; the one failure, always RACE×RELIG, was Mistral’s weakest too); a general demographic-geometry head recurs at a family-specific address (L33H9, fixed across types and checkpoints, scores  $\rho = 0.35$ –0.58 in five types, weakest again on RACE×RELIG); and ten billion tokens of simulation training barely move the map or the mouth (base→post-RL held-out fidelity shifts  $-0.07$  to  $+0.09$  with no systematic direction, output accuracy changes  $\leq 0.03$ , output-RDM fidelity improves 0.04)—consistent with the OdysSim authors’ own caution that better average-human simulation does not imply better subpopulation simulation.

The causal side does not yet replicate: patching (Sec. 5) has only been run on Mistral, so the map–use dissociation remains a one-model result.

Table 4: Causal patching results with the strong instrument. Ceiling = mean full-prompt-swap shift (Sec. 5.1, this instrument’s question set). Shift = mean directional shift toward the target group’s true distribution; % of ceiling in parentheses where meaningful.  $p_{\text{item}}$  = Wilcoxon vs. random-head control over 240 items (descriptive; items are clustered);  $p_{\text{pair}}$  = exact sign-flip over the 12 pairs (primary).

| Type (split-half fidelity) | Ceiling | 32 heads @ L11       | Single L11 H16                                  | $p_{\text{item}}$ (32h) | $p_{\text{pair}}$ (32h) |
|----------------------------|---------|----------------------|-------------------------------------------------|-------------------------|-------------------------|
| AGE×POLPARTY (0.63)        | +0.0165 | +0.0067 (41%)        | −0.0001 (n.s.)                                  | 0.0027                  | 0.019                   |
| RELIG×POLPARTY (0.52)      | +0.0020 | +0.0034 (170%, n.s.) | −0.0004 (n.s.)                                  | 0.16                    | 0.15                    |
| RACE×RELIG (0.21)          | +0.0026 | +0.0017 (n.s.)       | −0.0005 (wrong dir., $p_{\text{pair}}=1/4096$ ) | 0.31                    | 0.29                    |

## 5 Results II: Whether the Faithful Geometry Is Used

### 5.1 Outputs are nearly identity-invariant under natural prompting

Before asking whether any internal location causally drives answers, we measure the ceiling: how much does the model’s predicted opinion distribution move when the *entire* demographic identity in the prompt is replaced? Averaged over pairs and questions, the answer is almost not at all:  $\text{WD}(\hat{p}_A, p_B^{\text{real}}) - \text{WD}(\hat{p}_B, p_B^{\text{real}}) \approx 0.004$  against a total prediction error of  $\sim 0.37$  ( $<2\%$ ). (One illustrative item, not an aggregate statistic: an 11-point real gap between 18–29- and 50–64-year-olds moves by only 1.7 points under a full persona swap— $\sim 15\%$  of the true difference.)

Worse, the small variation that does exist is directionally uninformative: the prediction for group  $A$  is *no closer* to  $A$ ’s own ground truth than to another group’s (median advantage  $\approx 0$ ; sign test  $\approx$  coin flip in 2 of 3 tested types). The model produces slightly-different-but-equally-wrong answers per persona.

We re-measured this ceiling on the instruction-tuned Mistral-7B-Instruct-v0.2 under its own chat template: the near-invariance replicates, with the instruct model’s full-identity swap moving predictions by 3.9% of its total error (Appendix C.3; Table 11). This near-invariance is a property of the model family, not of the raw-completion interface, though both estimates assume the letter-probability read-out (Sec. 8).

The aggregate also hides a large per-type spread that inverts the fidelity ranking at the two extremes (Appendix C.3).

### 5.2 Patching at layer 11: a real but small effect

Patching L11 H16 activations between group prompts (donor→base) at the *prediction* token, with 1–3 heads at natural scale, moves nothing (protocol in Appendix C.1). But the ceiling above shows this instrument had a dynamic range of under 2% of total prediction error to begin with—any causal effect this small would be undetectable regardless of whether it exists.

We therefore repeat the intervention with an instrument matched in strength to prior successful steering work [15]: all 32 heads of layer 11 are patched jointly at a single identity token, so the edit propagates to every later token (protocol in Appendix C.1). Across the three types tested here the result is **graded**, and the gradation appears to line up with each type’s fidelity from Sec. 4; Sec. 5.3 extends the experiment to all six types and shows that this alignment does *not* generalise.

For AGE×POLPARTY, jointly patching all 32 heads at layer 11 shifts predictions toward the target group’s true distribution: 59% of trials move in the correct direction (vs. 32% under an empirical random-head control), recovering 41% of the achievable ceiling. The primary, pair-level test is marginal ( $p_{\text{pair}} = 0.019$ ; adding layer 18: 0.016) and does not survive a Holm correction over

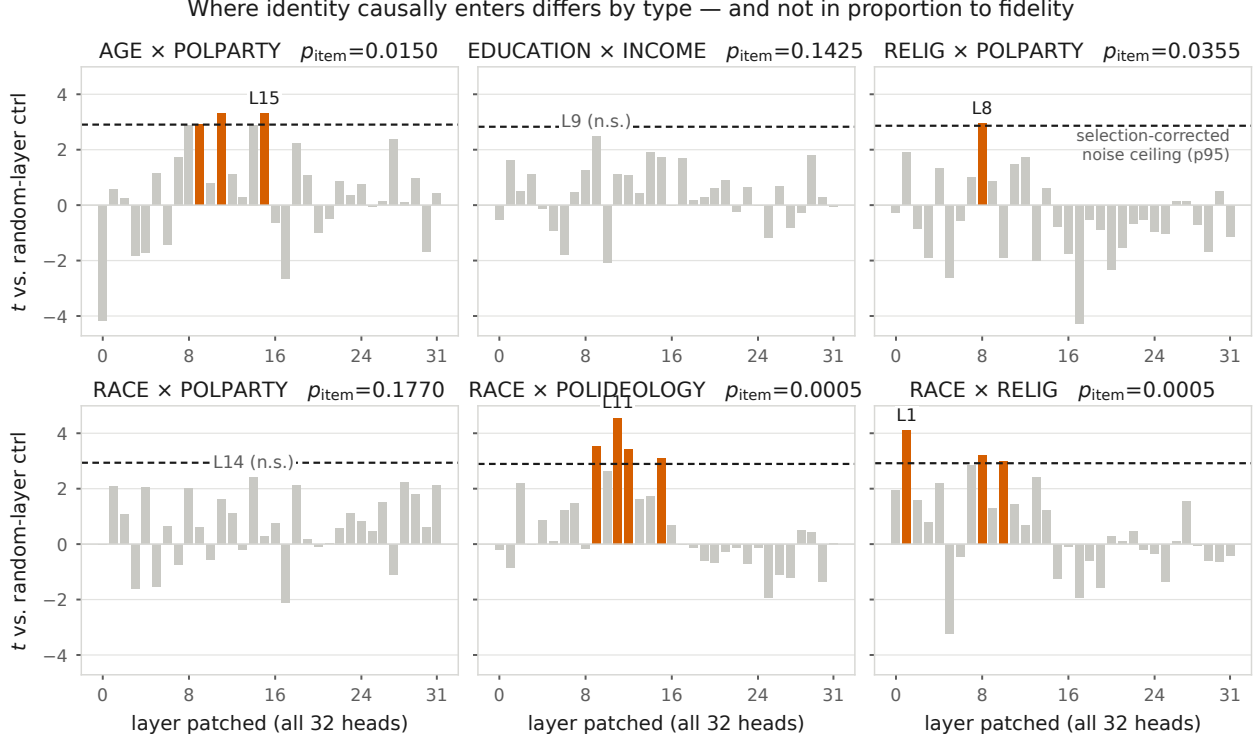


Figure 3: Causal effect of patching all 32 heads of each layer in turn, at the identity token ( $n=240$  pair-question items per type). *Descriptive* view: bars are the item-level paired  $t$  of the directional shift toward the target group’s true distribution against a random-layer control; the dashed line is the item-level max-statistic noise ceiling. Inference in the text and Table 5 is cluster-robust (sign flips at the level of the 12 pairs).

the family of 24 tests behind Table 4. Identity is thus only *plausibly* load-bearing here; Sec. 5.3 localises further.

For RACE×RELIG, the 32-head patch is not significant ( $p_{\text{pair}} = 0.29$ ), while patching L11 H16 *alone* shifts predictions in the *wrong* direction with the most consistent effect in the table (worse than the random-head control in all 12 pairs;  $p_{\text{pair}} = 1/4096$ ). A donor-control experiment shows this backfire requires a coherent identity donor rather than merely a large disruption to the representation (Appendix C.4).

Fidelity and single-unit causal sufficiency are dissociable: L11 H16 alone never produces a correct, significant shift in any type, and a correct effect requires the joint action of all 32 heads—faithful as a representation, but insufficient (or actively misleading) as a sole intervention target. Scaling the patch further backfires rather than strengthens it (Appendix C.4).

### 5.3 The causal locus does not track fidelity

Section 5.2 localises an effect at layer 11 for one type and finds none for another—too few types to tell whether causal use tracks fidelity at all. We therefore replace the hypothesis-driven probe with an exhaustive one, the same 32-head identity-token intervention applied at *every* layer, for all six attribute types (procedure and selection corrections in Appendix C.5).

Figure 3 shows the descriptive item-level picture; the pair-level statistics behind Table 5 are conservative (12 clusters per type) and prune it sharply. No type survives the pair-level max-

Table 5: Per-layer patching sweep, all six types. Fidelity = split-half median (one estimator used consistently; see text for sensitivity). Item-level columns are descriptive ( $t$  over 240 items; max-statistic  $p$  over 32 layers); the primary inference is the pair-level max-statistic  $p$  (sign flips over 12 pairs, exact) and the pair-level  $p$  at the a-priori locus L11 (no layer selection). The item-level and pair-level winning layers can differ, since averaging within pairs reweights the evidence. Output ceiling = mean shift from replacing the entire identity.

| Type             | Fidelity | item-level (descriptive) |      |            | pair-level (primary) |            |                             | Out. ceiling |
|------------------|----------|--------------------------|------|------------|----------------------|------------|-----------------------------|--------------|
|                  |          | Win layer                | $t$  | $p_{\max}$ | $t_{\text{win}}$     | $p_{\max}$ | $t@L11 \rightarrow p$       |              |
| RACE×POLIDEOLOGY | 0.30     | L11                      | 4.55 | 0.0005     | 3.22                 | 0.066      | <b>3.22</b> → <b>0.0020</b> | 0.0176       |
| RACE×RELIG       | 0.21     | L1                       | 4.10 | 0.0005     | 2.94 (L8)            | 0.122      | 0.68 → 0.27                 | 0.0026       |
| AGE×POLPARTY     | 0.63     | L15                      | 3.31 | 0.0150     | 2.37 (L14)           | 0.266      | 1.93 → 0.022                | 0.0165       |
| RELIG×POLPARTY   | 0.52     | L8                       | 2.94 | 0.0355     | 2.58 (L8)            | 0.210      | 1.11 → 0.15                 | 0.0020       |
| EDUCATION×INCOME | 0.58     | L9                       | 2.47 | 0.1425     | 1.57 (L15)           | 0.822      | 0.75 → 0.23                 | 0.0194       |
| RACE×POLPARTY    | 0.52     | L14                      | 2.40 | 0.1770     | 2.00 (L4)            | 0.524      | 1.28 → 0.11                 | 0.0032       |

Table 6: Fixed-location causal test, pair-level (exact sign-flip over 12 pairs, one-sided). L11 was fixed by Sec. 4 before any sweep; L1 is pre-registered only for the upper block (see text). Bonferroni for the 12 tests is 0.0042; the single bold entry survives it.

| Type                                                                 | $t @ L11$   | $p_{\text{pair}}$ | $t @ L1$ | $p_{\text{pair}}$ |
|----------------------------------------------------------------------|-------------|-------------------|----------|-------------------|
| <i>types not in the sweep that suggested L1:</i>                     |             |                   |          |                   |
| RACE×POLIDEOLOGY                                                     | <b>3.22</b> | <b>0.0020</b>     | −0.74    | 0.77              |
| RACE×POLPARTY                                                        | 1.28        | 0.114             | 1.97     | 0.033             |
| EDUCATION×INCOME                                                     | 0.75        | 0.234             | 1.24     | 0.123             |
| <i>types that were in that sweep (L1 column not selection-free):</i> |             |                   |          |                   |
| AGE×POLPARTY                                                         | 1.93        | 0.022             | 0.34     | 0.386             |
| RELIG×POLPARTY                                                       | 1.11        | 0.150             | 1.46     | 0.089             |
| RACE×RELIG                                                           | 0.68        | 0.267             | 2.46     | 0.016             |

statistic correction (best: RACE×POLIDEOLOGY,  $p = 0.066$ ). The a-priori test at L11 (no layer selection; fixed by Sec. 4) is cleaner: RACE×POLIDEOLOGY reaches  $t = 3.22$ , exact  $p = 0.0020$ , the only entry that also survives a Bonferroni correction over the 12 fixed-location tests below (Table 6); AGE×POLPARTY at L11 ( $p = 0.022$ ) is suggestive but not conclusive. The item-level loci otherwise cluster mid-stack and at layer 1 (selection caveat in Appendix C.5).

The natural hypothesis after Sec. 5.2—that causal strength should track fidelity—does not survive the extension to six types: the rank correlation between fidelity and pair-level causal strength is nowhere significantly positive under any estimator we have, including the lexical-partial fidelity of Sec. 4.3 (+0.26 to −0.77 depending on ruler and layer, all  $p \geq 0.07$ ; weakly powered at  $n=6$ , Figure 4). What carries the weight is the two extreme types, tested *directly* rather than by comparing significance verdicts [11]: the top-fidelity types—whichever estimator names them—show no correction-surviving single-layer locus (EDUCATION×INCOME has the largest output ceiling and largest raw shift of any type, yet  $p = 0.82$  at its best layer; AGE×POLPARTY, the split-half leader, reaches only a suggestive  $p = 0.022$  at L11 that fails every correction here), while RACE×POLIDEOLOGY—among the weakest maps under every estimator—is the single result that survives everything ( $t = 3.22$  at L11, exact  $p = 0.0020$ , surviving Bonferroni, recovering 42% of its ceiling). Representational fidelity is therefore not what decides whether identity is causally

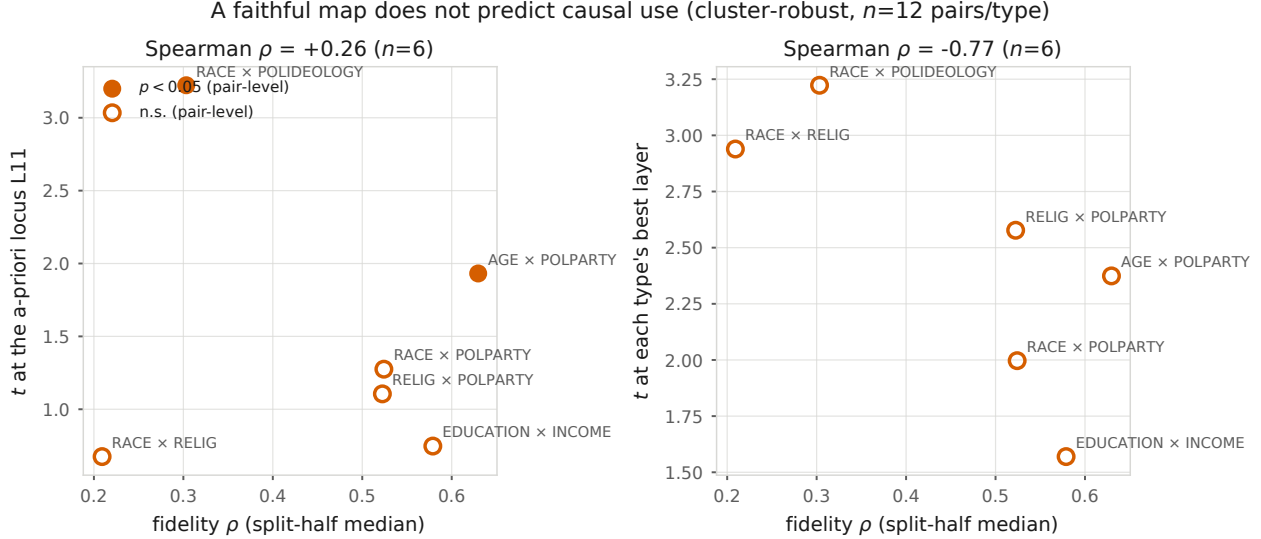


Figure 4: Representational fidelity (split-half median) against cluster-robust causal strength (pair-level  $t$ ), one point per attribute type. Left: at the a-priori locus L11 (filled = exact pair-level  $p < 0.05$ ). Right: at each type’s own best layer (filled = survives the pair-level max-statistic correction; none does).

used: the clearest causal pathway sits in the type with the poorest map, and the graded pattern of Sec. 5.2 does not generalise to all six types.

Sec. 4’s fidelity is scored on an identity-only prompt, whereas the causal and probe experiments are scored in a full QA context, so a tempting alternative reading is that the geometry simply does not survive into the answering context, not that the model fails to consult its map. Measured directly in that context—on the same activations collected for the probe (Sec. 5.4)—the map does weaken in five of six types, but the dissociation itself persists under the QA-context ruler at the same a-priori locus ( $\rho = +0.03$ ,  $p = 0.96$ ), so “we measured the wrong map” does not explain the dissociation away (Appendix C.6).

#### 5.4 Reading the map directly: probe versus output

The causal experiments ask whether the model *uses* what it encodes; the complementary question is how much there is to use if we read the faithful location directly rather than listen to the model’s mouth. We fit a ridge probe, evaluated leave-one-cell-out, at three locations—full layer 11, the single head L11 H16, and layer 1—against two references: the model’s own answer, and a group-blind baseline built from other cells’ real answers on the same question (full protocol in Appendices D.2, D.1).

In all six types the probe is substantially closer to survey truth than the model’s own answer (22–30% closer overall, better on 65–68% of individual items), not merely by better calibration: fitting a temperature parameter to the mouth barely helps ( $0.300 \rightarrow 0.298$  for AGE×POLPARTY), reproducing against real survey ground truth the claim that internal states are more informative than generated answers [15].

But the gain is question-level, not group-level: a group-blind baseline that ignores who is being asked and predicts the mean real distribution of the other cells on the same question beats every read-out we have, by a wide margin (0.133 vs. 0.211 for AGE×POLPARTY; 0.135 vs. 0.234 for

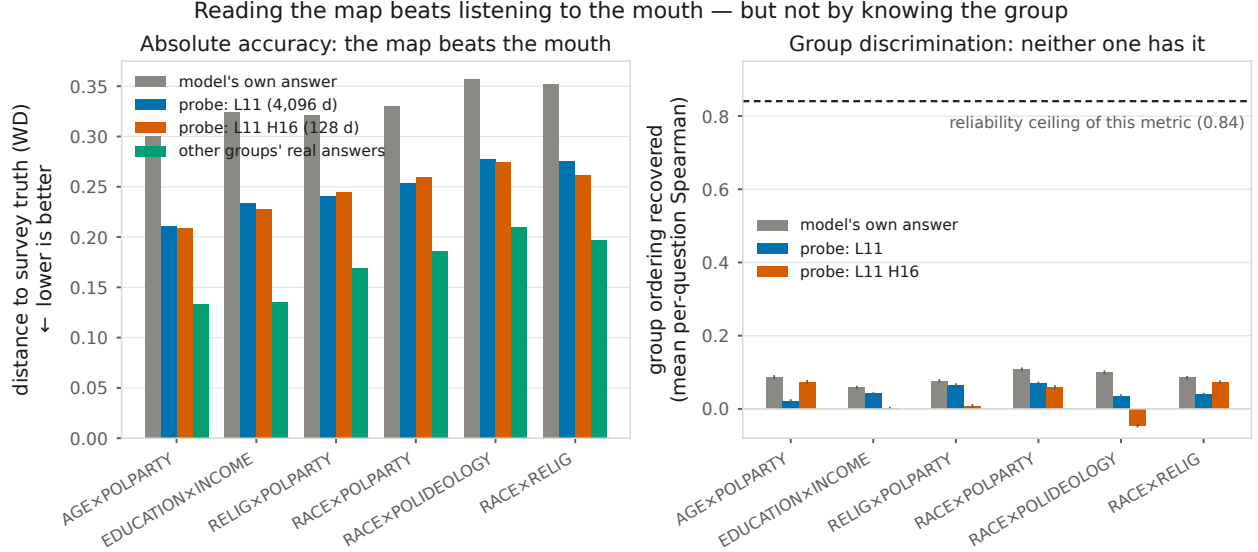


Figure 5: Left: distance to survey truth (Wasserstein, lower is better) for the model’s own answer, probes read from the faithful location, and a group-blind baseline built from other cells’ real answers. Right: how much of the true per-question group *ordering* each read-out recovers (mean per-question Spearman across cells, half-split evaluation; error bars are standard errors), against the reliability ceiling of the metric given survey sampling noise.

EDUCATION×INCOME). Ranking cells within each question by predicted opinion and correlating with the true ranking (half-split, to avoid a leave-one-out artifact) gives a flat null for everything: the full-layer L11 probe recovers only +0.02 to +0.07, the single head L11 H16 is noisier still (−0.05 to +0.07), and the model’s own answers recover +0.06 to +0.11—at least as much as either probe—against a reliability ceiling of +0.81 to +0.86.

These two results measure different resolutions and are consistent: Sec. 4 scores group geometry aggregated over hundreds of questions, where the arrangement mirrors survey structure ( $\rho$  up to 0.63 held-out); this section scores per-item discrimination, where the signal is near zero. The model holds a group map that is faithful on average, but not sharp enough to say who answers what on any particular question.

If the goal is a per-group opinion distribution, reading 128 dimensions of one attention head is markedly better than asking the model, but still worse than using other groups’ real answers to the same question (practical implications in Sec. 6).

## 6 Discussion

**Three properties, three literatures.** “Can LLMs simulate populations?” conflates three claims. Prior work established property (i), *readable* [15], and documented the behavioral symptom from outside (38, 37). We establish property (ii), *faithfully arranged*—at specific locations, unevenly across attribute types—and show that property (iii), *causally used*, is not a function of it. Under cluster-robust inference, causal use is not absent—one type shows a clear, Bonferroni-surviving pathway at L11, and two more are suggestive—but it does not follow the map (Sec. 5.3). Natural-prompt simulation therefore does not fail simply because the map is weak; a good map is no guarantee the model consults it.

**A mechanistic candidate for the Sim2Real gap—and a warning about explaining it by representation quality.** Behavioral homogeneity of LLM simulators is well measured from outside [38]. Our results offer a consistent inside account: outputs vary little across personas overall (Sec. 5.1), and forcing identity through the network moves them only slightly under a deliberately strong instrument. What our results do *not* support is that simulation fails because the internal representation is poor: homogeneity looks like at least two mechanisms—insufficient causal throughput where the map is good, and no good map to throughput where it is not. One qualification: the behavioral literature evaluates post-trained assistants, whereas we measure a base model; since post-training is itself a plausible cause of homogenization, our results establish only that a representational component of homogeneity predates post-training, rather than explaining any particular assistant’s behavior.

**Why surface fine-tuning may not transfer.** SubPOP-style fine-tuning [29] improves in-distribution accuracy but degrades on unseen surveys. If faithful structure already exists internally, output-surface fine-tuning may re-teach what the model already encodes—a survey-specific surface mapping instead of a connection to the existing map. This is a testable hypothesis, not a demonstrated mechanism; the direct test (does reading from the faithful location transfer where surface fine-tuning does not?) is future work.

**Practical guidance now.** For population simulation: check second-order fidelity before relating groups, treat demographic attributes as heterogeneous (Sec. 4), and measure identity conditioning rather than assume it. **Do not steer through a single apparently-faithful unit without a corruption control:** our causal analysis’s most consistent effect is a *harmful* one from doing exactly this (RACE×RELIG, L11 H16 alone, wrong direction,  $p_{\text{pair}} = 1/4096$ ; Sec. 5.2). Read a per-group distribution rather than ask for it: a linear probe on L11 H16 alone lands 21–31% closer to survey truth than the model’s own answers (Sec. 5.4), yet a group-blind baseline still wins on per-question ordering.

## 7 Conclusion

We asked whether a model that can be asked about a demographic group also faithfully represents, and actually uses, that group’s opinion structure. The answer dissociates. Beyond established readability [15], the group geometry is *faithfully arranged* at specific locations—selection-corrected  $\rho$  up to 0.63—but only weakly *causally used*: a full identity swap moves predictions by under 2% of their error, and a linear probe on the faithful location lands 21–31% closer to survey truth than the model’s own answers, yet recovers almost no per-question ordering of groups. Readable, faithfully arranged, and causally used are therefore separable properties of the same model, and treating them as one claim is why “can LLMs simulate populations?” has stayed unresolved: a model can encode a faithful map of a population and still not consult it when asked. Settling how general this dissociation is will need cross-model causal replication and cross-institution ground truth beyond a single survey house (Sec. 8).

## 8 Limitations

**Scope of replication.** The representational findings replicate on three checkpoints of a second model family (Sec. 4.6), but the patching experiments—and the map-use dissociation with them—have so far only run on Mistral-7B; output-side near-invariance does replicate on an instruction-

tuned Mistral (Sec. 5.1), so it is not a base-model artifact. Every patch is single-layer, so redundant encoding across layers remains an alternative explanation for a null result only partially ruled out (Sec. 5.3, caveat v). L11 H16 itself was first noticed in per-type top-10 lists before being fixed for the all-type test; Sec. 4.6 supplies a second-model check (L33 H9, selection-free on five types), but a preregistered replication is still outstanding.

**Measurement interface and contexts.** Output distributions are read from option-letter logits after **Answer:**—the standard silicon-sampling interface, but known to diverge from free-text answers [35]. Our output-side claims (Sec. 5) therefore scope to this interface: identity barely reaches the letter-probability read-out, not necessarily free-text generation, which we did not measure; the representation-side claims (Sec. 4) do not depend on the read-out at all. Relatedly, the two sides are measured in different input contexts (identity-only vs. QA), and the map is measurably weaker in the QA context (Sec. 5.3)—the fidelity-causality correlation is unchanged there, but our headline fidelity numbers describe an identity-only prompt and should be read as such.

**Ground truth and sample.** All distances are measured against one US-centric institution, Pew ATP; cross-institution transfer (e.g., scoring the map against GSS-derived inter-group distances) and cross-country replication (e.g., WVS, with a known language confound) are future work. The dissociation claim rests on six attribute types—a purposive sample of the 66 possible pairs among our twelve attributes (Sec. 3, Appendix A.1)—enough for two unambiguous counterexamples but not to generalize to intersections outside this set, and some cells have thin survey samples, with NaN pairs excluded (Appendix A.1). With cluster-robust inference the causal experiments have twelve effective units per type: enough for one clear result and direct between-type contrasts, not to certify a sweep winner against 31 rivals or estimate the fidelity-causality relationship precisely ( $n=6$  types); scaling pairs, not questions, is the binding constraint for follow-up work.

**Correlational map, one lexical encoder.** RSA fidelity is correlational; the patching program (Sec. 5.2, Sec. 5.3) addresses use, not the map’s origin. The partial-fidelity control (Sec. 4.3) uses one small sentence encoder as the lexical baseline; static word vectors cannot represent several value strings (e.g. age ranges), so a stronger lexical model would only make the control stricter.

## Ethics Statement

Simulating the opinions of demographic groups is dual-use: it supports legitimate survey-methodology research—pilot design, non-response modelling, coverage checks—but the same capability can manufacture apparently representative public opinion, or reinforce stereotypes by presenting a model’s guess about a group as that group’s view.

Our findings cut against the permissive reading: current LLM outputs are *not* substitutes for human respondents. Under natural prompting the model’s answers move by less than 2% of their error when the entire demographic identity is replaced, and the small movement that exists is not reliably toward the target group’s truth. The failure is also unequal—race-based attributes are the weakest and most prompt-fragile everywhere measured, so simulated outputs for racial groups are least trustworthy where misuse would be most harmful. We therefore caution against policy-adjacent uses of simulated group opinions.

Interpretability results of this kind carry their own risk: a faithful location is a natural target for steering, and Sec. 5.2 shows naive steering through such a location can push predictions *away* from the truth in low-fidelity types—read the map before trusting it as a lever.

All data are public, aggregated Pew ATP response distributions with no personally identifiable information; no human subjects were recruited for this work.

## Reproducibility Statement

All experiments use the public checkpoint `mistralai/Mistral-7B-v0.1` (fp16), prompts and pre-processing fully specified in Appendix A.1–A.3, and fixed seeds (42 throughout; permutation analyses additionally state their counts). Ground truth derives from public, aggregated Pew American Trends Panel response distributions in the OpinionQA/SubPOP format [25, 29]; no individual-level data are redistributed. GPU work ran on freely available 2×T4 instances; all statistics and figures regenerate from released CSVs via included scripts (analysis code and a figure generator accompany the repository, which we will release publicly upon publication).

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| Attribute type   | Cells | Pairs |
|------------------|-------|-------|
| AGE×POLPARTY     | 18    | 153   |
| EDUCATION×INCOME | 33    | 528   |
| RACE×POLIDEOLOGY | 30    | 435   |
| RACE×RELIG       | 30    | 435   |
| RELIG×POLPARTY   | 30    | 435   |
| RACE×POLPARTY    | 28    | 378   |

Table 7: Cell and same-type pair counts per attribute type.  $Pairs = \binom{cells}{2}$ , the number of RSA observations available for that type.

[38] Xuhui Zhou, Weiwei Sun, Qianou Ma, Yiqing Xie, Jiarui Liu, Weihua Du, Sean Welleck, Yiming Yang, Graham Neubig, Sherry Tongshuang Wu, and Maarten Sap. Mind the sim2real gap in user simulation for agentic tasks. *arXiv preprint arXiv:2603.11245*, 2026.

## A Data and Prompts

### A.1 Data preprocessing and cell inventory

We use the intersectional OpinionQA table derived from Pew American Trends Panel microdata [25, 29]: 177,217 group–question rows covering 1,494 questions from 15 ATP waves (26, 27, 29, 32, 34, 36, 41, 42, 43, 45, 49, 50, 54, 82, 92). Each row carries a group’s response distribution over that question’s ordinal scale; the option list is truncated to the ordinal length, dropping the trailing *Refused* category.

A *cell* is one intersection of two attribute values within one attribute type. The six types contribute 169 cells in total (Table 7):

The survey-side distance  $D_{ij}^{\text{real}}$  averages the Wasserstein distance over questions answered by *both* cells, capped at 300 questions sampled per pair (seed 42); questions whose ordinal scale differs in length between the two cells are skipped. Pairs with no shared questions stay NaN and are excluded pairwise from RSA; the selection-correction analyses (Sec. 3.5) additionally drop whole cells that produce NaN pairs, which removes 5–6 cells in the race-containing types and shifts their  $\rho$  values by  $\lesssim 0.04$ . The patching experiments further restrict to questions with at most six options (so the answer letters fit A–F) and to pairs sharing at least 20 such questions. Median cell sample size is  $n_{\text{unweighted}} = 132$  (minimum 30).

### A.2 Why these six attribute pairs

Sec. 3 explains in one sentence why these six pairs were chosen; the full rationale and a data-richness check follow.

Two (RACE×RELIG, RACE×POLPARTY) directly instantiate a dimension the closest prior work names as an open question: how to “ensure that LLMs faithfully capture opinions along other dimensions that are not explored in this work, such as intersections of demographic identities” [29, p. 9]—in our cell set, e.g., *White Mormon*, *Asian Republican*. The remaining four were chosen to span distinct social/political axes rather than for data richness.

This choice was not, however, biased toward the best-populated pairs, as the following ranking check shows. The twelve demographic attributes in this corpus admit  $\binom{12}{2} = 66$  possible attribute

| Attribute type (used) | Cells | Rank of 66 (tied block) |
|-----------------------|-------|-------------------------|
| EDUCATION×INCOME      | 33    | 7th (7–8)               |
| RACE×POLIDEOLOGY      | 30    | 18th (18–22)            |
| RELIG×POLPARTY        | 30    | 18th (18–22)            |
| RACE×RELIG            | 30    | 18th (18–22)            |
| RACE×POLPARTY         | 28    | 23rd (23–24)            |
| AGE×POLPARTY          | 18    | 44th (44–46)            |

Table 8: Rank of each used pair among the 66 possible attribute pairs by number of cells clearing  $n \geq 30$ . Ranks are the best rank within each tied block (competition ranking); the parenthesised range gives the positions that block occupies.

pairs. Applying the same per-wave  $n \geq 30$  rule used above to all 66—counting, for each pair, the number of distinct attribute-value combinations that ever clear the threshold in any of the 15 waves—ranks our six pairs 7th to 44th of 66 by cell count (Table 8). The richest pair by this measure (INCOME×RELIG, 39 cells) and the poorest (CITIZEN×SEX, 5 cells) were never tested. Selection was therefore not biased toward the best-populated pairs; the full ranking script and output accompany the repository.

### A.3 Prompt templates (verbatim)

For attribute type  $t$  with human-readable attribute labels  $(\ell_1, \ell_2)$  and cell values  $(v_1, v_2)$ , the four templates are:

**T0 (third-person declarative).** This survey respondent’s  $\ell_1$  is  $v_1$  and their  $\ell_2$  is  $v_2$ .

**T1 (first-person).** I am answering this survey. My  $\ell_1$  is  $v_1$  and my  $\ell_2$  is  $v_2$ .

**T2 (structured profile).** Survey respondent profile:

–  $\ell_1$ :  $v_1$

–  $\ell_2$ :  $v_2$

This respondent is about to answer opinion questions.

**T3 (QA format).** Question: What is this survey respondent’s  $\ell_1$  and  $\ell_2$ ?

Answer: Their  $\ell_1$  is  $v_1$  and their  $\ell_2$  is  $v_2$ .

The label pairs are: race/religion, race/political party affiliation, race/political ideology, religion/political party affiliation, highest level of education/household income, age group/political party affiliation. TMEAN averages the four extracted representations (not the four prompts). Example instantiation of T0 for one cell: This survey respondent’s age group is 30–49 and their political party affiliation is Democrat.

## B Fidelity-Map Details

### B.1 Anisotropy and mean-centering

The last-token residual stream is strongly anisotropic: the ratio  $\|\bar{v}\|/\|\bar{v}\|$  over the 169 cell embeddings is 0.950 (it would be  $\approx 0$  for randomly oriented vectors and 1 if all vectors were identical in direction). At layer 17 the pairwise cosine distances have median 0.10 and maximum 0.27; the *same-type* pairs, which are the ones the task must separate, are the tightest (maximum 0.12).

Table 9: Fidelity ( $\rho$  vs. survey truth) at layer 17 before and after removing the shared direction ( $v_i \leftarrow v_i - \bar{v}$ ). Centering was run on three of the six types; full detail above.

| Read-out     | Pooled        | AGE×POLPARTY                             | EDUCATION×INCOME | RACE×RELIG |
|--------------|---------------|------------------------------------------|------------------|------------|
|              | (mixes types) | within-type (the regime used throughout) |                  |            |
| raw          | +0.185        | +0.455                                   | +0.490           | +0.072     |
| mean-centred | +0.301        | +0.375                                   | +0.454           | +0.033     |
| change       | <b>+0.116</b> | −0.080                                   | −0.036           | −0.039     |

Table 10: Fidelity averaged over all 32 heads within each quarter of the stack (multi-cue TMEAN read-out). Unlike Figure 1, which takes the best head per layer, a block mean involves no selection, so these values are comparable across depths. Bold marks each type’s peak block.

| Type             | L0–7 | L8–15       | L16–23      | L24–31      |
|------------------|------|-------------|-------------|-------------|
| AGE×POLPARTY     | 0.36 | <b>0.47</b> | 0.42        | 0.42        |
| EDUCATION×INCOME | 0.43 | <b>0.53</b> | 0.46        | 0.48        |
| RELIG×POLPARTY   | 0.13 | 0.21        | <b>0.23</b> | 0.22        |
| RACE×POLPARTY    | 0.17 | <b>0.24</b> | 0.24        | 0.22        |
| RACE×POLIDEOLOGY | 0.18 | <b>0.27</b> | 0.22        | 0.21        |
| RACE×RELIG       | 0.09 | 0.12        | 0.15        | <b>0.17</b> |

Removing the shared direction ( $v_i \leftarrow v_i - \bar{v}$ ) nearly doubles the pooled correlation (+0.185  $\rightarrow$  +0.301 at layer 17, consistently +0.07 to +0.16 across all 32 layers), and the residual variance is genuinely multidimensional afterwards (PC1 31%, PC2 20%, PC3 12%). We nevertheless do *not* adopt centering, because the gain does not survive the decomposition that matters here: within type—the only regime used for fidelity scoring and for the  $w_{ij}$  kernel—centering *lowers*  $\rho$  (AGE×POLPARTY 0.455  $\rightarrow$  0.375; EDUCATION×INCOME 0.490  $\rightarrow$  0.454; RACE×RELIG 0.072  $\rightarrow$  0.033). The pooled improvement is an artifact of aligning scale *between* types, not sharpening signal *within* them. Table 9 summarizes this within-type comparison.

## B.2 Full fidelity map

The table behind Figures 6 and 1 (`peta_kesetiaan_full.csv`) holds 32,670 rows: 1,089 locations  $\times$  5 template conditions  $\times$  6 types. The layer-0 residual read-out is NaN by construction—the final token is shared across cells, so that representation is constant—which is why the residual curves start at layer 1.

## B.3 Selection-correction details

All three corrections run on the extracted representations, no GPU required (`notebooks/10_koreksi_seleksi_peta_local.ipynb`).

**Max-statistic permutation** (`cek2`, 2,000 permutations): cell labels of the survey RDM are permuted, the whole 1,024-head map is recomputed, and the *maximum*  $\rho$  is retained, giving the distribution of what a champion selected from noise can reach. Null means are 0.21–0.32 and null 95th percentiles 0.34–0.46; the observed maxima exceed them in every type. This test runs on the NaN-cleaned cell subset, on which the three race-type maxima are slightly higher than the full-set values in Table 1 (0.45  $\rightarrow$  0.49, 0.50  $\rightarrow$  0.52, 0.61  $\rightarrow$  0.65); null and observation are computed on the same subset, so the comparison is internally consistent.

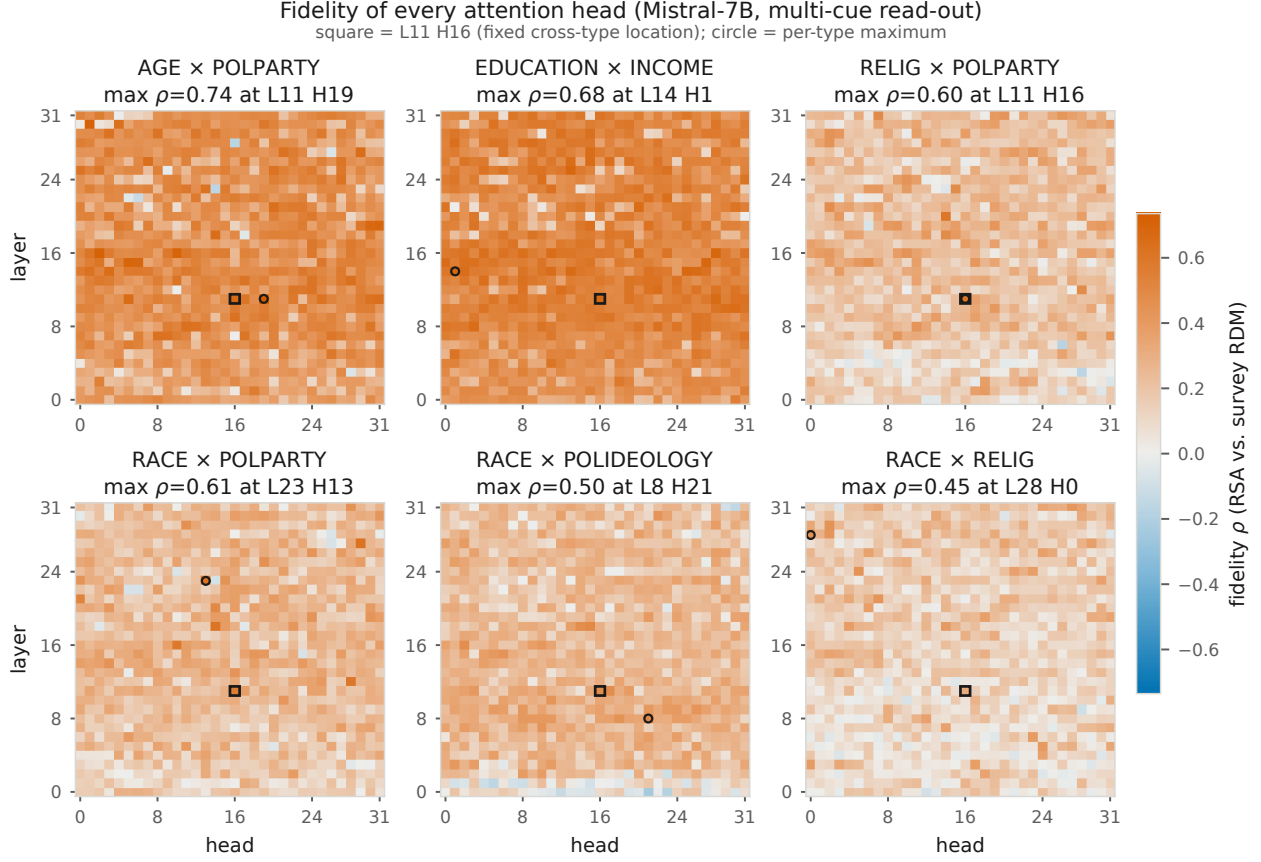


Figure 6: The full fidelity map: every attention head (32 layers × 32 heads) scored against survey ground truth, per attribute type, multi-cue read-out. Square marks the fixed location L11 H16; circle marks that type’s own maximum; the underlying table is described above. In the two strongest types 84–92% of the 1,024 heads exceed  $\rho = 0.3$ , against 5% in RACE×RELIG; the level varies with depth rather than at random (Table 10 above) and falls for race-containing types (bottom row). Maxima printed here are uncorrected and match the *selected* column of Table 1; see that table for held-out values.

**Held-out template (cek1):** the head is selected on three templates and scored on the fourth, rotating over all four folds.

**Split-half (cek3, 200 resamples):** cells are split at random, the head is selected on one half and scored on the other; we report the median. The same table records how often each head is re-selected—L11 H16 in 87/200 splits for RELIG×POLPARTY versus a maximum of 22/200 for any head in RACE×RELIG, the quantitative form of the “head choice is a lottery” statement in Sec. 4.

**Fixed-location test (cek4):** L11 H16 and L18 H14 are fixed in advance and scored per type with a per-location permutation test (1,000 permutations) and a conservative ×1024 Bonferroni threshold. L11 H16 passes in all six types; L18 H14 fails only in RACE×RELIG ( $p_{\text{raw}} = 0.004$ ).

## B.4 Random-projection control for the head–residual comparison

The three candidate families (1,024 heads; 33 residual layers; 1,024 random 128-dimensional Gaussian projections of the residual stream, 32 per layer, fixed across folds) are put through one identi-

cal held-out-template selection pipeline. Held-out fidelity per type (head / residual / projection): AGE 0.52/0.39/0.42; EDU 0.54/0.51/0.49; RELIG×PP 0.44/0.17/0.18; RACE×PP 0.47/0.28/0.30; RACE×PI 0.38/0.26/0.29; RACE×RELIG 0.29/0.16/0.17. (This pipeline’s held-out variant selects on the mean of three templates and is slightly more conservative than the one behind Table 1; comparisons are valid within this table only.)

## B.5 Lexical-baseline control (partial correlations)

The lexical RDM embeds either the two attribute-value phrases (averaged) or the full T0 identity sentence with all-MiniLM-L6-v2 [34] and takes cosine distances. Partial Spearman is computed on ranks; significance is by cell-label permutation of the survey RDM (2,000 permutations), holding both model and lexical RDMs fixed. Per type (value-phrase variant): lexical-vs-survey  $\rho$  = AGE 0.39, EDU 0.64, RELIG×PP 0.21, RACE×PP 0.21, RACE×PI 0.31, RACE×RELIG 0.28; L11 H16’s partial fidelity = 0.59, 0.35, 0.59, 0.58, 0.37 (all  $p < 0.0005$ ) and 0.23 ( $p = 0.056$ ) respectively. The full-sentence variant gives the same qualitative picture with higher lexical baselines (up to 0.64 for AGE).

## B.6 Reliability ceiling for the fidelity map

Survey-side reliability simulates two independent respondent samples per (cell, question) at the real unweighted sample sizes and correlates the two resulting RDMs (10 repetitions): 0.985–0.994 per type. Model-side reliability correlates RDMs built from disjoint template halves (all three 2v2 splits, averaged): 0.72–0.94 at L11 H16. The attenuation ceiling  $\sqrt{r_{\text{survey}} r_{\text{model}}}$  is 0.84–0.96; observed fidelity reaches 39–74% of it.

## B.7 Sensitivity of the survey-side ruler

Normalising each question’s WD by its scale range changes L11 H16’s fidelity by at most 0.03 in any type (e.g., AGE 0.68 → 0.69; EDU 0.66 → 0.63). Restricting to questions answered by *every* cell of a type is possible for three types (77–102 common questions): fidelity holds for AGE (0.68 → 0.64) and RELIG×POLPARTY (0.61 → 0.56; best head 0.72), and drops for EDUCATION×INCOME (0.66 → 0.44). For the three race-containing types no question is answered by all cells, so this variant is undefined there.

# C Causal-Experiment Details

## C.1 Patching protocol details

**Read/write point.** All interventions are applied to the input of `self_attn.o_proj`—the per-head activations before the out-projection mixes them—so a head is addressed as a 128-dimensional slice of the 4,096-dimensional vector. The patch is an interpolation toward the donor,

$$x \leftarrow x + \alpha(d - x), \tag{3}$$

with  $\alpha = 1$  (full replacement) and  $\alpha = 5$  (amplification) as the two reported settings,  $d$  the donor group’s activation at the corresponding position.

**v1 (Sec. 5, weak instrument).** Identity is a natural sentence, the patch is applied at the final (prediction) token only, and 1–3 heads are patched at natural scale.

**v2 (Sec. 5.2, strong instrument).** Identity is localized to the answer token of a QA-formatted demographic block (Question: What is this survey respondent’s ... / A.

... / **Answer:** B), and the patch is applied at those identity-answer token positions, so the edit propagates to every later token. Conditions: `patch_L11_all132` (all 32 heads at layer 11), `patch_L11H16` (the star head alone), `patch_L11L18_all` (both layers), `patch_L11_all132_x5` ( $\alpha = 5$ ), and `patch_randhead` (a single randomly drawn head, excluded from {L11 H16, L18 H14, L11 H19}, fixed for the run) as the control. A `self_patch` sanity condition—patching a cell with its own activations, which must produce zero shift—is run on the first three pairs of each type. Scope: 3 types  $\times$  12 pairs  $\times$  20 questions, the questions chosen per pair by largest real inter-group Wasserstein distance, giving  $n = 240$  per type-condition cell.

**Sweep (Sec. 5.3).** The v2 all-32-heads intervention is repeated at every layer, for all six attribute types (three in the first sweep, three added later with identical settings: 12 pairs  $\times$  20 maximum-disagreement questions,  $n = 240$  items). The selection correction is a sign-flip permutation test (2,000 permutations) in which the same random sign is applied to an item across all layers, preserving the between-layer correlation structure; the statistic is the maximum  $t$  over the 32 layers. All six types were corrected in one run with one seed; re-running that code on the first three types reproduces the earlier figures to within permutation noise (e.g.  $t = 4.098$  vs. 4.10 for RACE $\times$ RELIG at layer 1).

**Read-out.** The predicted opinion distribution is the softmax over the first  $n_{\text{opt}}$  answer-letter token ids at the position after **Answer:**, and is compared to the target group’s real distribution by Wasserstein distance on the question’s ordinal scale. The reported shift is  $\text{WD}(\hat{p}_A, p_B^{\text{real}}) - \text{WD}(\hat{p}_{\text{patched}}, p_B^{\text{real}})$ : positive means the patch moved the prediction toward the donor group’s *truth*, not merely toward the model’s own prediction for that group.

**Donor controls** (for the L11 H16 backfire, Sec. 5.2). On the same 12 pairs and 20 questions per type (RACE $\times$ RELIG and, as a negative control, AGE $\times$ POLPARTY), the single-head patch is repeated under four donors captured at the same positions: (i) the paired group’s activation (replication of the original condition); (ii–iii) the activation of two other same-type identities, drawn at random per pair excluding both members; and (iv) a dimension-shuffled donor—the paired group’s vector with its 128 head dimensions permuted by a fixed per-pair permutation, matching location, norm, and coordinate statistics while destroying structure. Donors are captured per (cell, question) exactly as in the main experiment. All inference is pair-level (exact sign-flip, two-sided). Raw output movement is tracked as  $\text{WD}(\hat{p}_{\text{patched}}, \hat{p}_A)$  so that harm can be compared against perturbation size across conditions.

## C.2 Cluster-robust causal inference

For every test we first average within pair (12 values), then enumerate all  $2^{12} = 4,096$  sign-flip combinations exactly; max-statistic variants take the maximum  $t$  over the 32 layers within each flip.

## C.3 Identity-swap ceiling: instruction-tuned model and per-type spread

**Instruction-tuned model.** We re-measured the identity-swap ceiling on Mistral-7B-Instruct-v0.2 under its own chat template, with the same cells and questions (40 per type, all same-type cell pairs; under this estimator the base model’s swap movement is 3.2% of its total error, rather than the  $<2\%$  above, which is computed on the original question set). The near-invariance replicates: the instruct model’s full-identity swap moves predictions by 3.9% of its total error. What chat tuning changes is accuracy and direction, not identity-dependence: absolute error nearly doubles (mean WD 0.66 vs. 0.37), while the small movement becomes reliably directionally correct (group A’s prediction closer to A’s own truth than to B’s in 64% of pairs; sign test  $p < 0.001$  in all six

Table 11: Identity-swap ceiling per attribute type, base vs. instruction-tuned Mistral (40 questions per type, all same-type cell pairs). *Swap* = movement of the predicted distribution under a full identity replacement, as a percentage of that type’s total prediction error. WD = mean absolute error to survey truth (lower is better). *Dir.* = fraction of pairs where group *A*’s prediction is closer to *A*’s own truth than to *B*’s, with the sign-test *p*; 0.50 is a coin flip.

| Type             | Swap (% of error) |        | Mean WD |        | Dir. ( <i>p</i> ) |              |
|------------------|-------------------|--------|---------|--------|-------------------|--------------|
|                  | base              | instr. | base    | instr. | base              | instr.       |
| AGE×POLPARTY     | 2.6               | 3.4    | 0.36    | 0.65   | 0.56 (.045)       | 0.75 (<.001) |
| EDUCATION×INCOME | 1.0               | 2.4    | 0.33    | 0.59   | 0.55 (.002)       | 0.62 (<.001) |
| RELIG×POLPARTY   | 1.6               | 3.4    | 0.37    | 0.64   | 0.52 (.293)       | 0.65 (<.001) |
| RACE×POLPARTY    | 3.6               | 1.5    | 0.35    | 0.70   | 0.54 (.061)       | 0.59 (<.001) |
| RACE×POLIDEOLOGY | 9.2               | 9.9    | 0.37    | 0.66   | 0.62 (<.001)      | 0.64 (<.001) |
| RACE×RELIG       | 1.2               | 2.9    | 0.46    | 0.70   | 0.52 (.297)       | 0.59 (<.001) |

types, vs. three of six at  $p < 0.05$ , one at  $p < 0.001$ , for the base model). An instruction-tuned model is thus *on average* slightly more identity-responsive and substantially less accurate; the effect is not uniform, and one type moves the other way (RACE×POLPARTY, 3.6% → 1.5%; Table 11). The near-invariance itself is a property of the model family, not of the raw-completion interface, though both estimates assume the letter-probability read-out (Sec. 8).

**Per-type spread.** Per type (Table 11), the identity swap moves RACE×POLIDEOLOGY by 9.2% of its error but EDUCATION×INCOME by only 1.0%. First, the ranking is not the fidelity ranking: the two extremes of *fidelity*—RELIG×POLPARTY, among the most faithful types (Table 1), and RACE×POLIDEOLOGY, mid-table—sit at 1.6% and 9.2% respectively, i.e. the more faithful type is the *less* identity-responsive one. The same inversion holds under the independent ceiling estimator of Table 5, computed on the causal item set: RELIG×POLPARTY is last of six there (0.0020) and RACE×POLIDEOLOGY second (0.0176). Second, the two estimators do *not* agree in general—EDUCATION×INCOME is last here and first there (0.0194), reflecting different question sets and absolute vs. error-normalised scaling. The spread therefore supports the fidelity–use inversion at the two ends, but not a per-type ordering.

## C.4 Amplification and donor controls

For RACE×RELIG, replacing 128 dimensions mid-computation is a destructive intervention, so we ran a donor-control experiment on the same pairs and questions (independent run; the effect itself replicates, pair-level  $t = -2.4$ , exact  $p = 0.005$ ). A *dimension-shuffled* donor—the target group’s own vector with its 128 head dimensions permuted, so the location, norm, and coordinate statistics match while the structure is destroyed—perturbs the output *twice* as strongly in raw movement, yet produces no systematic drift at all (pair-level  $t = -0.1$ ): if harm merely scaled with disruption, this condition would be the most harmful, and it is the least. The consistent away-from-truth drift appears only with a *coherent identity donor*. Whether it further requires the *paired* group’s donor is suggestive but unsettled: donors from other same-type identities trend the same way weakly (each n.s.), and the direct paired contrast against their mean reaches  $t = -2.0$ , exact  $p = 0.073$ . Narrowly, then: single-unit intervention at an apparently faithful location remains unreliable in direction.

Scaling the patch by  $5\times$  (matching the amplification levels used in SAE-feature steering, 15)

makes the shift *negative* in all three types (condition not shown in Table 4; full results in Appendix C.1). Linear extrapolation in raw activation space likely pushes representations off-manifold, unlike steering in a constrained SAE feature space—amplifying a raw signal is not a free strengthening operation.

### C.5 Layer-sweep procedure, control sensitivity, and caveats

**Procedure.** For each pair-question item we assemble a single batch of 33 copies of the prompt: one patched at each of the 32 layers, plus one patched at a randomly chosen layer as a control, so all conditions see identical inputs and one forward pass answers the whole sweep ( $n=240$  pair-question items per type). The statistic per layer is the paired  $t$  of its directional shift against the random-layer control. Two selection problems require correction. First, the winner is selected from 32 candidate layers, so we apply a max-statistic permutation correction [22], with the same sign flip applied across all layers to preserve the between-layer correlation structure. Second, and more consequentially, the 240 items are *clustered* into 12 pairs (Sec. 3.5), so all primary  $p$ -values below flip signs at the *pair* level: per-pair mean shifts, all  $2^{12}$  flip combinations enumerated exactly. Item-level statistics are shown as descriptive detail only—treating the 240 items as independent understates the  $p$ -values several-fold.

One further wrinkle affects the layer-1 reading specifically: layer 1 was noticed as a candidate locus only during the first sweep, which included RACE×RELIG, so for that one type the layer-1 column of Table 6 is not selection-free (its pair-level  $p = 0.016$  should be read with that caveat, and the “early-layer hub” reading stays a hypothesis for the cross-model replication); for the three types added later, both the L11 and L1 columns are genuinely fixed in advance.

**Control sensitivity.** The random-layer control can itself land on a causally relevant layer, deflating measured effects; and “shift” is defined against the unpatched baseline prediction, so a no-patch reference exists implicitly. Re-running every pair-level test against zero instead of against the control changes no conclusion (e.g., RACE×POLIDEOLOGY@L11:  $t = 3.24$ ,  $p = 0.0022$  vs. zero; full table in the released analysis code).

**Five caveats.** (i) Absolute effects are small throughout (0.001–0.012), commensurate with output ceilings that are themselves small; the surviving effect is statistically robust, not practically large. (ii) Most loci were *discovered* by the sweep rather than predicted, so they warrant replication in a second model (Sec. 4.6)—though the L11 column of Table 6 is free of per-type selection. (iii) With six types, any cross-type correlation is weakly powered; our claim rests on the two counterexamples, not on the correlation coefficient. (iv) What layer 1 encodes mechanistically remains open: a check on already-collected residual streams was inconclusive because the layer-0 read-out is degenerate in our prompt design (the final token is shared across cells), so we cannot contrast layer 1 against pure embeddings; layers 1 and 2 are near-identical in geometry ( $\rho = 0.90$ ) and both differ appreciably from layer 11 ( $\rho = 0.78$ ); see Sec. 8. (v) A single-layer patch may fail to detect a causally-used representation that is redundantly encoded across many layers—the pattern a high, broadly-distributed fidelity (Figure 6) would predict. Our one direct check argues against strong compensation, though only for one type: AGE×POLPARTY’s two-layer patch (L11+L18 jointly, Sec. 5.2) moves the pair-level effect only marginally beyond the single-layer result ( $p = 0.019 \rightarrow 0.016$ ). EDUCATION×INCOME—the type this caveat matters most for—has not been tested under a multi-layer patch; redundant, single-layer-invisible use remains an alternative we have not ruled out for it.

**Magnitude versus consistency.** Three separate times in this sweep the layer with the largest raw shift fails correction while a smaller, steadier layer passes: RELIG×POLPARTY (L12/L11 largest, L8 passes), RACE×RELIG (L7 +0.0030 and L10 +0.0026 fail, L1 passes at +0.0010, with the selection-inflation caveat given above for this type), and EDUCATION×INCOME (largest shift in the paper, nothing passes at either clustering level). Reporting mean effect sizes without a selection-and-cluster-corrected variance test would have produced three wrong localisations here.

## C.6 The fidelity map in the QA context

Sec. 4’s fidelity is scored on an identity-only prompt (no opinion question, last token), whereas the causal and probe experiments are scored in a full QA context at the opinion-answer position. A tempting alternative reading of the dissociation is therefore not “the model does not consult its map” but “the faithful geometry is not the one that exists once a question is present.” We test this on the activations already collected for the probe (Sec. 5.4), holding question composition fixed across cells—each cell is averaged over the same question set, the questions answered by every cell of its type, since coverage is only  $\geq 60\%$  per cell and unmatched subsets would themselves contribute inter-cell distance. The resulting RDM is near-perfectly reliable (split-half  $\rho = 0.98\text{--}0.99$  in the five types with enough all-cell-shared questions to split, above the identity-only side’s  $0.72\text{--}0.94$ ), so what follows is a property of the representation, not of measurement noise.

**The map does degrade:** in five of six types QA-context fidelity falls well below the identity-only value (EDUCATION×INCOME  $0.67 \rightarrow 0.10$ ; RACE×POLPARTY  $0.59 \rightarrow 0.17$ ; RELIG×POLPARTY  $0.60 \rightarrow 0.23$ ; RACE×POLIDEOLOGY  $0.47 \rightarrow 0.24$ ; AGE×POLPARTY  $0.67 \rightarrow 0.39$ ), the exception being RACE×RELIG, which *rises* ( $0.33 \rightarrow 0.49$ ). Sec. 4 therefore characterises the geometry of an identity-only prompt, and only a weaker version of that geometry survives into the context where the model actually answers. **The dissociation, however, does not depend on the ruler:** with the QA-context ruler at the a-priori locus L11 the correlation is  $\rho = +0.03$  ( $p = 0.96$ ), the same null the identity-only ruler gives. The type with the clearest causal locus (RACE×POLIDEOLOGY) is mid-pack on the QA ruler too, and the most faithful type under it (RACE×RELIG) has no detectable L11 effect ( $p = 0.27$ ). At each type’s own *selected* best layer the correlation is ruler-unstable ( $-0.77$  identity-only vs.  $+0.71$  QA-context, both n.s.), which is the same caution we already attach to that quantity. Measured in the very context where the causal experiments look for it, fidelity still fails to predict causal use—so “we measured the wrong map” does not explain the dissociation away.

## D Probe Details

### D.1 Probe protocol and the group-ordering test

**Data.** Questions are selected for *coverage* rather than disagreement: within each attribute type we keep questions answered by at least 60% of that type’s cells (minimum 10), then draw a random sample of 250 of them (seed 42). Random sampling matters: the highest-coverage questions concentrate in two or three survey waves, which would restrict the probe to a narrow topic range. The resulting set is 33,395 (cell, question) rows, 4,051–7,309 per type, spanning up to all 15 ATP waves per type. Activations are captured at the opinion-answer position and stored in half precision.

**Fitting.** Features are reduced with PCA (128 components, fit on training rows only) and a ridge regression predicts the six-dimensional padded response distribution; predictions are truncated to the question’s option count and renormalised. Evaluation is leave-one-cell-out: every prediction is for a demographic cell absent from training. The ridge penalty is chosen inside each training fold

by a further split over cells (median selected value  $\alpha = 1$ ). Scores are mean Wasserstein distance to the real distribution on the question’s ordinal scale.

**References.** “Mouth” is the softmax over answer-letter tokens from the same forward pass. The group-blind baseline predicts, for each held-out row, the mean real distribution of the *other* cells on that question (training cells only).

**Group-ordering test.** Correlations are Spearman. The leave-one-out artifact the half-split design of Sec. 5.4 avoids is severe: leave-one-out predictions are anti-correlated with the held-out value by construction, driving the group-blind baseline to  $\rho \approx -0.99$ .

**Reliability ceiling.** To calibrate the group-ordering numbers we simulate two independent survey samples per cell (multinomial draws at that cell’s own unweighted sample size) and run the same test between them. This gives +0.81 to +0.86 per type, i.e. the maximum a perfect read-out could achieve given survey sampling noise alone.

## D.2 Probe fitting procedure

For every (cell, question) pair we collect the activation at the opinion-answer position and fit a ridge probe (128 PCA components) to predict the group’s *real* response distribution (full protocol in Appendix D.1). Evaluation is leave-one-cell-out: the probe is always scored on a demographic cell it never saw, and the regularisation strength is chosen inside the training fold. We read three locations—the full layer 11 (4,096 dims), the single head L11 H16 (128 dims), and layer 1 (4,096 dims)—and compare against two references: the model’s own answer distribution (“the mouth”), and a group-blind baseline that predicts the mean real distribution of the *other* cells on that question. Broad-coverage questions are used throughout (250 questions per type, each answered by at least 60% of that type’s cells; 33,395 rows in total), which is what the earlier, narrower question set could not support.

## E The Failed Consistency-Loss Pilot

Sec. 4.1 is motivated by a downstream failure, reported here for completeness. We had implemented a group-consistency auxiliary loss  $L_{\text{group}}$  that pulls the model’s predictions for demographically similar cells together, weighting pairs by a kernel  $w_{ij}$  built from last-token residual-stream embeddings. The pilot failed; the diagnosis:

1.  $L_{\text{group}}$  performed no better than a plain probe, and both lost to simple shrinkage; it was insensitive to  $N$ .
2. The embeddings of all 26 RACE×RELIG cells occupied one narrow cone (all pairwise similarities  $> 0.91$ ).
3. A *random* probe already drove  $L_{\text{group}}$  to  $\approx 0$ —the failure was geometric, not a training problem.
4. The highest-weighted neighbours under  $w_{ij}$  were frequently not the cells with the most similar real answers.
5. Raising  $\lambda$  from 1 to 100 monotonically worsened accuracy.
6. A single shared probe let 23 data-rich cells dominate the sparse target cell.

Points 2–4 are the direct motivation for asking where a *better* read-out might live, which is this paper’s question.